NumPy Full Notes & Practice Sheet

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1. **Introduction to NumPy**
2. Powerful Python library for numerical computing.
3. Supports arrays, matrices, mathematical functions.
4. Faster than lists: contiguous memory, C implementation, vectorized operations.

```
import numpy as np
arr = np.array([1,2,3])
print(arr)
```

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1. **Array Creation Methods**
2. `np.array()`, `np.zeros()`, `np.ones()`, `np.full()`
3. `np.arange()`, `np.linspace()`, `np.eye()`, `np.identity()`
4. `np.random.rand()`, `np.random.randint()`

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1. **Array Properties**
2. `ndim`, `shape`, `size`, `dtype`, `itemsize`, `nbytes`

```
arr = np.array([[1,2,3],[4,5,6]])
print(arr.ndim, arr.shape, arr.size, arr.dtype)
```

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1. **Reshaping & Manipulation**
2. `reshape()`, `flatten()`, `ravel()`, `transpose()`, `resize()`

```
arr = np.arange(1,10)
arr2d = arr.reshape(3,3)
print(arr2d.flatten(), arr2d.ravel(), arr2d.T)
```

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1. **Mathematical Operations**
2. `add()`, `subtract()`, `multiply()`, `divide()`, `power()`
3. `sin()`, `cos()`, `tan()`
4. `round()`, `floor()`, `ceil()`

```
a = np.array([1,2,3])
b = np.array([4,5,6])
print(np.add(a,b))
```

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1. **Statistical Functions**
2. `sum()`, `mean()`, `median()`, `std()`, `var()`, `min()`, `max()`, `argmin()`, `argmax()`

```
arr = np.array([1,2,3,4,5,6])
print(np.mean(arr), np.median(arr), np.std(arr))
```

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1. **Searching & Sorting**
2. `sort()`, `argsort()`, `where()`, `unique()`, `searchsorted()`

```
arr = np.array([4,2,5,1,3])
print(np.sort(arr), np.argsort(arr), np.where(arr>3), np.unique(arr))
```

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1. **Linear Algebra**
2. `dot()`, `matmul()`, `linalg.inv()`, `linalg.det()`, `linalg.solve()`, `linalg.eig()`

```
A = np.array([[1,2],[3,4]])
b = np.array([5,11])
x = np.linalg.solve(A,b)
print(x)
```

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1. **Practice Exercises**
2. 3 x 3 matrix 1 - 9 : `np.arange(1,10).reshape(3,3)`
3. Mean, median, std: `np.mean(arr), np.median(arr), np.std(arr)`
4. Reshape 1 D → 2 D: `arr.reshape(3,4)`
5. Random integers > 5 0 : `rand[rand>50]`
6. Solve linear equation: `np.linalg.solve(A,b)`

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1. **Interview Questions**
2. Why is NumPy faster than Python lists?
3. Difference between `reshape()` and `resize()`
4. Difference between `flatten()` and `ravel()`
5. What is broadcasting?

6 . Explain `axis` parameter.